Do Green Investments Enhance UK Productivity?

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0.1 Non-technical summary

The United Kingdom's productivity has seen minimal growth since 2008, a period marked by secular stagnation. This report delves into the relationship between green investments, particularly those aimed at achieving net-zero emissions, and their impact on regional productivity within the UK. It leverages data from the UK's Net Zero Innovation Portfolio, focusing on the Net Zero Innovation Programme (NZIP) and its predecessor, the Energy Innovation Programme (EIP), which together have channeled over £2 billion into local green technology projects over the past decade.

Key Findings:

- **Green Investments and Productivity**: The report finds an *observational* increase in labour productivity in local authorities that benefited from NZIP funding of between £0.20-£0.50 per hour worked, with productivity gains manifesting 3-4 years after the investment. This suggests a potential positive impact of green investments on productivity, albeit with a delayed effect.
- **Challenges in Establishing Causality**: Establishing a direct causal link between green investments and productivity enhancements is complex due to data limitations, potential biases, and the inherent challenges in measuring the impact of R&D investments.
- **Heterogeneity in Impact**: The analysis indicates that the impact of green investments on productivity is heterogeneous, varying by sector, region, and time period. It suggests that regions with lower initial productivity levels may benefit more from green investments, with potential short-term negative impacts but positive long-term gains.
- **Policy Implications**: The findings highlight the dual benefits of green investments in addressing the UK's productivity problem and achieving net-zero emissions by 2050. They underscore the importance of targeting investments in low-productivity regions and promoting long-term investment strategies.

Recommendations for Future Research:

- **Broadening the Analysis**: Future studies could enrich the analysis by incorporating additional productivity measures, drawing international comparisons, and applying more refined spatial methodologies to account for spillover effects.
- **Robust Causal Identification**: Identifying sources of exogenous variation, such as using near misses in investment funding for a regression discontinuity approach, could provide a more robust causal identification of the effects of green investments on productivity.
- **Data Availability**: The report emphasises the need for the government to make available high-quality, granular datasets on large-scale investment plans to facilitate more nuanced analyses and inform policy decisions.

1 Introduction

The UK is in a period of secular stagnation, with productivity barely rising since 2008. This worrying trend has been at the forefront of economists' research on how to restore productivity and growth to the UK. However, one theory behind the slowdown is the greater importance of transforming economies from polluting to net zero emissions which has prevented effective markets from operating efficiently.

Over the past decade green technologies have significantly developed leading to solar and wind being cheaper than fossil fuels in electricity production and since the COVID-19 pandemic a common narrative was to "Build Back Better" by boosting green investment to help restore economies. Such policies have been adopted in the US under the Inflation Reduction Act and Infrastructure Investment and Jobs Act and in the EU under the Green Deal Industrial Plan.

But literature on the impacts of green investment on productivity is not vast, especially when focused on the UK. When the government is seemingly split between economic growth and its aim to hit Net Zero emissions by 2050 such a lack of literature can inhibit key investment that could achieve both results.

Therefore, this paper aims to reduce such deficiencies by using the UK's Net Zero Innovation Portfolio to determine the place based effects of green investment on regional productivity

For our analysis we have taken data from two of the UK's flagship green investment programmes, the Net Zero Innovation Programme (NZIP) and its predecessor, the Energy Innovation Programme (EIP). Combined the programmes have delivered more than £2bn, over 10 years, investment at local levels to promote commercialising/development of green technologies. Successful projects are published online allowing us to extract and compile an experimental dataset of treatment at Local Authority (LA) and ONS regional levels.

The measure of productivity we have selected is Gross Value Added (GVA) per hour worked at the LA and ONS regional level. GVA is a commonly used productivity metric, primarily used to measure labour productivity.

We estimate a difference-in-difference model to identify the observational effect of NZIP funding on Local Productivity.

2 Literature Review

The evolving landscape of the net zero transition has led to vast uncertainties around the true impacts of green policies. This has created a prominent role for economic theory in decision-making which typically has seen net zero as a drag on productivity and growth.

This theoretical basis is highlighted by Ajayi and Pollitt (2022) who argue that maintaining total factor productivity (TFP) will be an ever increasing challenge as economies transform, this is in part due to the natural capital required in driving growth in an economy. When measures such as GDP (which TFP can explain when decomposed) exclude factors such as natural capital or improvements in health, the conflicting goals between using resources to drive growth and ensuring a sustainable use of resources come to a head. Furthermore, a move from increasing material throughputs to an altered composition of GDP (with resources used more efficiently) would represent a new steady state. Such a steady state would be in conflict with continuous GDP growth due to the priority to minimise throughputs and production. Ajayi and Pollitt (2022) also state the altered consumption of resources is not the only source of lowered growth. They quote the International Energy Agency in saying green industries, excluding solar, actually create the least jobs in manufacturing and thus further suggests lower production in the economy.

The authors also conduct some simple modelling using the National Grid ESO's Future Energy Scenarios (2021). They find energy efficiency is an important source of Net Zero but argue this is likely to further hinder growth. Investing in low-carbon solutions creates an increasing cost of inputs which their modelling suggests will be challenging in the 2020s. However, they do reveal productivity increases may come in the long-run as electrification increases and fossil fuels decline but in hard-to-abate energy and energy intensive sectors such as hydrogen there are likely further challenges, this view is support by another working paper which suggests that low productivity in the energy and energy intensive sectors since 2008 has partly been due to increased climate ambition (Ajayi, 2020).

However, Ajayi and Pollitt's work contains some pitfalls and fails to empirically test their assumptions. For example, costs of capital have been assumed constant which contradicts the trends associated with solar and wind energy which have significantly fallen over the past decade (EEIST, 2023), furthermore, their belief that economic growth and resource depletion come hand-in-hand is in contradiction to the rising level of intangible capital and services within growth (McKinsey, 2021). Finally, the lack of causal evidence in conclusions means the findings are insufficient to determine green investment causes lower productivity.

Turning to empirical evidence, Woo, C., Chun, D., Han, S., & Lee, D. (2013) use the Korea Innovation Survey to conduct a difference in difference based on firms that stated they took part in green innovation between 2007-2009. Using a dummy variable to distinguish between groups the authors find that labour productivity was higher for those who engaged in green innovation creating benefits to customers and firms. However this differed between firm sizes and industries where large firms in pollution intensive industries tended to invest more in environmental technology. This contradicts the theory within the working paper of Ajayi and Pollitt (2022) suggesting positive impacts. However, this paper is unable to establish causality which creates limitations in its conclusions.

Extending these findings We, Chen and Li (2023) analyse causal impacts of the Low-Carbon City Pilots in China and specifically focus on energy-intensive industries. As mentioned, these industries are likely to be the most affected by the costs of investment due to their naturally higher consumption of energy that would need to be decarbonised. By pilots only being available in certain regions this enables the authors to conduct a quasi-experimental analysis of green investment on productivity. Using a difference-in-difference test We, Chen and Li (2023) found that TFP was higher at the 10% significance level in regions with the LCCP policy compared to those without and this finding remains when looking specifically at energy intensive industries. This contradicts Ajayi and Pollitt (2022) but supports Woo, C., Chun, D., Han, S., & Lee, D. (2013). However, it is worth noting the impact was found to be greater on State Owned Enterprises than non and it was found that the impact was lower for areas with less human capital. This means applicability to the UK for using Net Zero as a way to increase productivity is limited due to the much higher representation of non-state enterprises and the government objective to level-up lower productivity regions which have lower human capital.

Moving towards evidence more representative of the UK's economy, Kalantzis and Niczyporuk (2021) analyse labour productivity improvements from energy efficiency measures. Under classical theory investing in energy efficiency measures presents an opportunity cost to other more effective investments which have been undertaken purely for Net Zero related reasons. However, using a survey of 15,000 firms in the EU27 and UK between 2018-2019 a causal link between energy efficiency has been identified. Using both a Probit-2SLS (two-stage least squares) and Direct-2SLS, the authors not only determine that investment in energy efficiency is productivity increasing but also that the amount of investment also causes a larger increase in productivity benefits. The paper pins this down to the non-energy benefits which impede rational decisions of businesses in looking at future cash flows causing underinvestment. Instead, the requirement to hit net zero enforces investment which actually causes a further gain to business. This paper significantly supports the view that green investment does raise productivity but due to focusing purely on energy efficiency and not looking into the heterogeneous impacts, it is limited in comparison to other papers.

Furthermore, Siedschlag and Yan (2022) conduct analysis on Ireland – a member of the EU – and conduct a more micro level assessment of green investment on wider variables and heterogenous drivers. By using difference in difference propensity score matching related to output growth, labour productivity, export intensity and energy intensity the paper significantly supports other empirical literature to suggest green investments improve performance. They also find that effects are largest for firms who are larger, foreign-owned, more productive and in lower tech industries. For other firms the effect is less significant and can be negative in the short-term but positive in the long-term.

This therefore supports our empirical evidence base to suggest that impacts are overall positive but that effects are highly heterogeneous. This suggests that in contradiction to Ajayi and Pollitt (2022) Net Zero actually provides an opportunity to improve productivity and UK growth. This can be due to a range of reasons such as firms not considering non-energy benefits or composition of energy sectors in an economy.

This indicates that Green Investment may be a lever in helping to improve the UK's productivity problem while also helping in the UK's goals to hit Net Zero emissions by 2050 but due to the lack of empirical evidence and the likely difference between populations further analysis is required.

3 Definitions

Green investment is a widely used term across media, government, academia and industry, however is ill-defined and without a common shared definition. We adopt a definition set out by the IMF (Eyraud et al., 2011): "investment necessary to reduce greenhouse gas and air pollutant emissions, without significantly reducing the production and consumption of non-energy goods.".

Productivity is better defined in literature and has been the subject of many studies due to its links with economic growth. We adopt a definition set out by the OECD: "Productivity is commonly defined as a ratio between the output volume and the volume of inputs." (Krugman, 1994).

Gross Value Added (GVA) is a commonly used output metric for productivity in UK government analysis (UK Gov, 2014; DEFRA, 2022). We take the definition of GVA from the ONS labour productivity statistics for this analysis: "An estimate of the volume of goods and services produced after subtracting the volume of intermediate goods and services used in the production process" (ONS, 2023). By dividing GVA (output) by hours worked in an area we can define a measure of productivity.

4 Data

4.1 ONS: Gross Value Added

Our productivity measure is taken from the latest Experimental Statistics publication on Subregional productivity in the UK (ONS, 2023). This breaks down GVA per hour worked at Local Authority Level by year.

4.2 NZIP & EIP: Data Overview

Data for GI comes from two UK programmes, the Net Zero Innovation Programme (NZIP) and the Energy Innovation Programme (EIP). NZIP "provides funding for low carbon technologies and systems" (DESNZ and BEIS, 2023). NZIP will run from 2021 to 2025 and builds on its predecessor EIP which ran from 2015-2021 (DESNZ, 2023). NZIP has committed £1.3bn of spend across 10 priority themes to develop and commercialise green technologies (DESNZ, 2023).

We note here that we do not have full coverage of all GI in the UK. NZIP only covers a fraction of public sector investment and the data available does not provide any data on private sector investment. Walker (2023) suggests that the public sector investment in GI was around £10bn in 2020, suggesting that NZIP may only cover around 10% of public sector investment. Data is sparse on what private sector GI may be, total funding (private and public sector), suggested by (Vianez, 2023), was £27bn and £23bn in 2011 and 2022 respectively (likely an underestimate as reporting from private companies on investment is poor).

We gave two examples of major investment mechanisms excluded from this analysis. The UK infrastructure bank (public sector) has invested significantly in the UK GI over the past 10 years (UKIB, 2024). In 2012 founded the UK Green Investment Bank as a non-departmental body of the Department for Business, Energy and Industrial Strategy (BEIS), now a private sector organisation, which manages assets in £bns on GI (Green Investment Group, 2023).

For the reasons above, whilst NZIP does cover a significant proportion of public sector GI in the UK, it is liable to substantial effects from selection bias and measurement error. We will discuss this in more detail later in the paper.

4.3 NZIP & EIP: Data Extraction

Funded projects are listed on GOV.UK, from which we extract project information. However, we faced unexpected issues with our dataset, requiring assumptions and corrections for analysis readiness, summarised in Table 1.

Table 1: Data Corrections

Issue	Resolution/Description	Implication
Location Data Missing	Used company headquarters or Companies House address when no location provided.	Increases "headquarters" effect, misallocating spend location.
General Location	Created custom lookup tables for matching to Local Authority (LA) or Region.	Manual matching introduces potential errors in LA/Region allocation.
Start Date as Range	Earliest date used as start.	Affects timing accuracy of treatment effect.
Missing Data	Missing data noted from summary statistics, mainly EIP spend.	Skews early years' treatment, biasing groups towards 2021 and NZIP projects.

4.4 Data Limitations, Issues and Potential Biases

Due to the way we have extracted NZIP data and the processing necessary, it can only be described as a highly-experimental dataset. Comparing the extracted data to the official published aggregated statistics (DESNZ, 2023), Overall, we capture less than half of the expected spend over a vastly different distribution of region. Whilst our data is in an appropriate format to use for analysis, the above means that any conclusions or results may be spurious until further investigation is done on why major differences are seen.

Additionally, our data (and thus analysis using this data) is subject to a number of potential biases. Firstly, we have selected NZIP only for this analysis, excluding all other investments (green or otherwise). Hence our sample data and following analysis may be subject to a selection bias (Delgado-Rodríguez and Llorca, 2004). Secondly, due to the selection bias in our sample, there is a likely significant non-systematic measurement error in our data, as it does not fully capture public or associated private green investment in an area (Local Authority or Region) which will affect our regressive analytical methods (Stefanski, 2000). Thirdly, the headquarters effect occurs due to a company's registered headquarters located in one place but there are other sites where the funded R&D takes place. As we have had to, in some cases, assign a location based on a company's headquarters, our data is subject to this (UKRI, 2019). Lastly, our data processing has been subject to manual intervention and interpretation. Whilst we have attempted to be as systematic and automate the processing as much as possible, our dataset and analysis is liable for human error. Another issue to be aware of, is that a large proportion of NZIP funding is focused on research and development (R&D) to bring technologies to commercialisation. The literature tells us that measuring changes in productivity after investment into R&D is difficult as the return may not be immediate and over a longer time scale (Pappas and Remer, 1985).

5 Methodology

As outlined, NZIP is a regional grant funding program. This means some areas were "treated" whereas some were not. As such, a Difference-in-Difference framework could be used to identify the effect of the scheme on key outcome variables relative to places that were not treated. From the outset, it is important to note that this is a descriptive and not a causal investigation, we will explain in more detail later in the paper. This approach has been chosen in line with research done by Huwei, W., Shuai, C., & Chien-Chiang, L. (2023) who implemented a two-way fixed effects model (TWFE) with fixed effects at the year and company level.

However, since treatment for NZIP is staggered — where the treatment groups are treated at different time periods — this presents complexities in designing a straightforward event study. A few key studies have highlighted the possible inaccuracies that may arise when using a conventional TWFE regression estimator in staggered adoption scenarios. Key contributions in this area include the studies by Sun and Abraham (2021), Callaway and Sant'Anna (2020), and Goodman-Bacon (2021). These studies underscore the risks of potential distortions in results when employing TWFE methods in settings where treatment adoption occurs at different times. As such, we utilise an alternative estimation given by Sun and Abraham (2020). This estimation method utilises "interaction-weighted" (IW) estimators for estimating dynamic treatment effects. Importantly, these estimates are robust to heterogeneous treatment effects across cohorts.

This estimation technique relies on three assumptions:

Assumption 1: (Parallel Trends in Baseline Outcomes)

$$\forall s \neq t, E[Y_{i,t}^{\infty} - Y_{i,s}^{\infty} | E_i = e] \text{is the same for all} e \in supp(E_i) \tag{1}$$

Assumption 2: (No Anticipatory Behaviour Prior to Treatment)

$$E[Y_{i,e+\ell}^{\infty} - Y_{i,e+\ell}^{\infty} | E_i = e] \forall e \in supp(E_i) \text{ and all } \ell > 0 \tag{2}$$

Assumption 3: (Treatment Effect Homogeneity)

$$\text{For each relative period} \ell, CATT_{e,\ell} \text{ does not depend on cohort } e \text{ and is equal to } ATT_{\ell} \tag{3}$$

5.1 Model Specification

Following Sun and Abraham (2020), we specify our model as:

$$Y_{gt} = \alpha + \sum_{k=T_0}^{-1} \beta_k \times \text{treat}_{gk} + \sum_{k=0}^{T_1} \beta_k \times \text{treat}_{gk} + X_{gt}\Gamma + \phi_g + \gamma_t + \varepsilon_{gt} \tag{4}$$

- Y_{gt} is the dependent variable - *Productivity*
- $treat_{sk}$ is a dummy variable that equals 1 if the observation's periods relative to the groups first treated period is the same as k, else it is 0. (it is 0 for all groups not treated)
- T_0 and T_1 are the lowest and highest number of leads and lags around the treatment period.

- X are controls, in our case Total NZIP Funding.
- ϕ_g represents the fixed effects for the region (LAD), accounting for unobserved heterogeneity across regions.
- γ_t represents the fixed effects for the year, accounting for unobserved heterogeneity across time.
- ϵ_{it} is the error term for each observation.

When using this modelling specification, one of the time periods must be dropped to avoid perfect multicollinearity (as in most fixed-effects set-ups). We have chosen 2020.

Fixed effects models exploit variation within an entity (like a region) over time, rather than variation between entities. This is crucial when you believe that the primary source of information for estimating your model comes from changes within an entity, rather than differences between entities. For example, in our model, we're more interested in how changes over time within the same region affect productivity, rather than how different regions compare to each other.

By including fixed effects for regions and years, we aim to eliminate confounding from all time-invariant characteristics. This is especially important in observational studies where random assignment is not possible, and there could be many unobserved, time-invariant factors affecting the outcome.

> **Causality**
>
> It is important to note that this analysis does not aim to elicit causal estimates of the effect of NZIP funding on productivity. This is because the treatment effect in this case is highly endogenous. It is almost certainly the case that Local Authorities that are more productive are more likely to receive NZIP funding as the bids come from existing organisations undertaking research and development, usually in areas with already high productivity. Therefore, any observed effects are purely descriptive.

5.2 Statistical Tests

A key condition for difference-in-difference estimations is the Parallel trends assumption. This states that, without the treatment, the outcomes for both groups would have exhibited parallel trends. This indicates that the change of the outcome variable in the absence of treatment is unaffected by the allocation of treatment. We conduct a series of cohort specific event studies as used by Miller (2019). Annex A shows plotted regression coefficients on treatment leads and lags with the aim to demonstrate the groups had comparable outcome dynamics pre-treatment.

We conduct a Hausmann test to determine whether the unique errors (unobserved individual effects) in our panel data model are correlated with the regressors. If they are correlated, the fixed effects model is preferred because it can control for this correlation by allowing the intercept to vary across individuals. If not, the random effects model is more efficient and is the preferred choice. The results in Annex B show that our p-value is less than 0.05, therefore we reject the null hypothesis that the preferred model is the random effects model (i.e., the unique errors are not correlated with the regressors) and use a fixed-effects model. Furthermore, we can run a Lagrange Multiplier Test for time effects. The test aims to detect whether the variance of the error terms in our model varies across time, which would suggest that incorporating time effects (random effects) could improve the model's accuracy. Annex C shows the null hypothesis is rejected at the 1% level of significance, strongly suggesting that there are significant time effects, we therefore add year fixed effects to preferred specification.

We do not include a dummy variable to indicate the impact of COVID19 as we are using year fixed effects, these absorb all year-specific variations, including those that could be captured by the COVID19 dummy. Since the dummy is a function of the year (1 for 2020 and onwards, 0 otherwise), the year fixed effects already account for the variation that the dummy would explain. Furthermore, the analysis is meant to capture within-group variation over time and the COVID19 dummy does not vary within these groups (it changes for all groups at the same time), so it will not provide additional information beyond what is captured by the fixed effects. Another possibility is to investigate a series of lock down dummies, however our temporal granularity is yearly which would not allow for this.

We test for heteroskedasticity through a Breusch-Pagan test, with the null hypothesis that the variance of the errors is constant. In annex D we strongly reject this at the 1% level and confirm that our errors are heteroscedastic. As shown by the variation in treatment effect in our data section, it is possible that this variance could be different between our treated and untreated groups. Therefore we conduct a Goldfeld-Quandt test to see whether the variance of the errors increases or decreases as the value of our treatment indicator changes. Annex E shows a high p-value, suggesting that there is not enough statistical evidence to reject the null hypothesis of constant variance between the two segments. This means that, according to the Goldfeld-Quandt test, the assumption of homoscedasticity cannot be rejected for this model. In other words, the test does not provide evidence of heteroscedasticity based on the segmentation by the 'treat' variable.

To address issues highlighted in the tests and to mitigate against other concerns related to the application of panel data geographically, namely Cross-sectional dependence and autocorrelation[1], we choose to use Driscoll-Kraay (1998) standard errors. This approach is often applied when dealing with large panels, i.e., datasets with a large number of cross-sectional units (N) and time periods (T), especially when N is large relative to T. Or in Difference-in-Difference estimations and other panel data methodologies where the researcher is concerned about the presence of spatial and temporal dependence that other robust standard errors (e.g., White's standard errors) do not address. We have a large number of cross-sectional units (292) relative to our time period.

6 Results

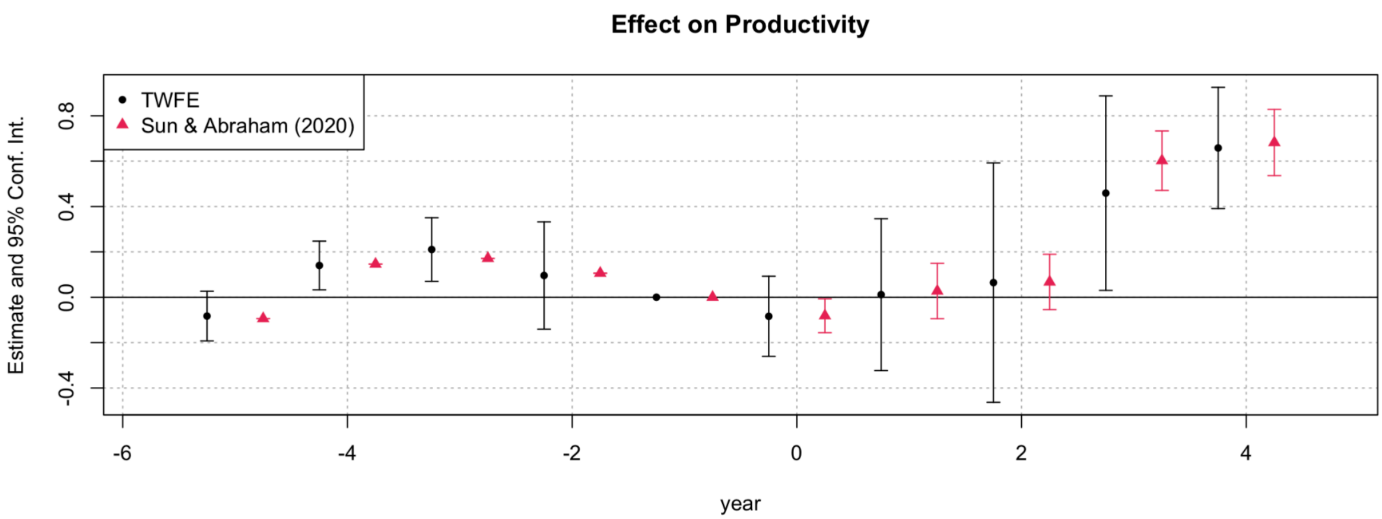

Figure 1: Difference-in-Difference estimation using OLS

Figure 1 shows the "Naive" TWFE alongside the Sun and Abraham (2020) estimates for the effect of treatment of NZIP funding relative to the cohort's treatment year. The estimates show effects near zero for the periods leading up to the treatment and for two years following this. Years 3 and 4 following the treatment show a statistically significant positive effect on productivity. This suggests a lag in positive treatment effects on productivity, which is to

[1]Productivity shocks or policy impacts in one area might spill over to neighbouring areas or even to distant areas with economic ties, leading to correlated error terms across cross-sections. Or, an area with high productivity in one year is likely to have relatively high productivity in the following year due to persistent factors like infrastructure, workforce skills, and local policies.

be expected as the treatment effect here is the distribution of the grant, as many of these projects are capital intensive and involve the construction of infrastructure, any positive effect on productivity is expected to occur several years after the grant has been given.

Annex F details the coefficient results for the SA estimator. For Total_Funding, the t-value is 0.572, and the p-value is 0.575, indicating that Funding is not statistically significant at conventional levels. This suggests that, holding other factors constant, changes in funding do not have a statistically significant association with productivity. The coefficients for the year fixed effects (relative to a reference year) show significant changes in productivity over time. The coefficients near zero earlier years (e.g., year::-5 to year::-1) suggest similar productivity levels compared to the reference year, while the positive coefficients for later years (e.g., year::3, year::4) indicate higher productivity levels. The significance of these coefficients ($p < 0.001$) suggests strong evidence of temporal variation in productivity. The coefficients for years closer to the reference year (year::0, year::1, year::2) are not statistically significant, indicating no substantial difference in productivity compared to the reference year. The significant positive coefficients for year::3 and year::4 (0.584 and 0.662, respectively) with very low p-values suggest a notable increase in productivity in these years compared to the reference year.

This, however, is only part of the picture. An important concept in difference-in-difference literature is the Average Treatment Effect on the Treated. This measures the effect of the intervention on those units that actually received the treatment. It contrasts with the Average Treatment Effect (ATE), which considers the effect of the treatment on the entire population, regardless of whether each unit received the treatment or not.

Importantly, we can decompose this ATT by cohort to investigate what the effect of NZIP was conditional on the year that the funding was given.

Table 2: ATT by Cohort

Variable	Coefficient	Standard Error
Dependent Var.:	**Productivity**	
Total_Funding	1.29e-8	(1.18e-8)
cohort = 2017	0.2693**	(0.0567)
cohort = 2018	0.5208***	(0.0474)
cohort = 2019	-0.3000***	(0.0432)
cohort = 2021	-0.1372***	(0.0163)
Fixed-Effects:		
lad_name	Yes	
year	Yes	
S.E. type:	Driscoll-Kra. (L=1)	
Observations:	1,752	
R2:	0.96727	
Within R2:	0.00359	
Signif. codes:	0.001 '***' 0.01 '**' 0.05 '*' 0.1 '.'	

This shows that there are clear differences in the ATT conditional on the years that the funding was given. For funding delivered in 2017 and 2018, there was an ATT of 0.269 and 0.521 respectively, which implies that on average, GVA per hour worked increased by £0.27 in Local Authorities that received treatment in 2017 and increased by £0.52 in 2018. However, funding that was delivered in 2019 and 2020 have opposite signs, values for these years indicate that GVA per hour decreased by £0.30 and £0.14 respectively. All results are significant at the 1% level.

6.1 Interpretation

The findings from our staggered difference-in-difference estimation reveal nuanced insights into the treatment effect of the NZIP grant scheme on Local Area GVA per hour worked. The temporal dynamics of the treatment effect underscore a delayed positive impact on productivity. This delay aligns with the nature of the NZIP funding, which primarily supports capital-intensive projects that necessitate time for construction and subsequent integration into local economies. The initial years post-treatment exhibit negligible changes in productivity, with statistically significant enhancements materialising only in the third and fourth years. This pattern corroborates the expectation that the fruits of infrastructure and innovation investments, such as those facilitated by the NZIP, typically manifest over a longer horizon and aligns with literature in Siedschlag and Yan (2022) who find benefits from green investment for high-technology accrued in the long-term.

The analysis of the SA estimation shows insignificance of the Total_Funding variable, suggesting that, in isolation, the magnitude of funding does not directly correlate with immediate productivity shifts. This finding might indicate that the effectiveness of the funding is contingent upon other factors, such as the nature of the projects funded, the efficiency of fund deployment, or other project specific characteristics.

The decomposition of the ATT by cohort provides a more granular perspective on the NZIP's efficacy. The differential impacts across cohorts suggest that the timing of funding receipt plays a critical role in determining the extent of productivity gains. The positive and significant ATT for the 2017 and 2018 cohorts indicates that funding allocated during these years effectively translated into productivity improvements. Conversely, the negative ATT for the 2019 and 2021 cohorts points to a detrimental effect on productivity. We posit that this is suggestive of either 1) Heterogenous project characteristics such as differences in the types of projects funded or inefficiencies in project implementation, not considered in our modelling, or 2) Exemplifying the expectation that any labour productivity benefits of infrastructure and innovation investments would typically manifest over a longer horizon.

Again, these findings cannot be interpreted as causal. Therefore, it's unclear whether any variations in productivity are directly due to the NZIP, as this merely reflects the changes in labour productivity in areas that also received NZIP funding.

6.2 Limitations

Aside from issues around endogenous treatment previously highlighted, we can evaluate to what extent our application meets the required assumptions for causal interpretation of SA estimation .

1) **Parallel Trends**, which requires that, in the absence of the portfolio, all local authorities would have followed similar productivity trends. This assumption is challenged by the inherent differences among local authorities, such as their economic structures and environmental priorities, which could affect their readiness and timing in adopting the portfolio's measures. Particularly, local authorities with a history of sustainability efforts might already be on a different productivity trajectory due to previous investments in green technologies. 2) **No Anticipatory Behaviour** requires that local authorities or businesses would not exhibit changes in anticipation of the portfolio's implementation. However, this assumption could be invalidated if local authorities or businesses have foresight or access to private information about the portfolio's interventions, leading them to adjust their strategies or operations preemptively. Such anticipatory behaviour could confound the observed effects of the portfolio on labour productivity. 3) **Treatment Effect Homogeneity** requires that the portfolio's impact might be uniform across local authorities, regardless of when they are introduced to the portfolio. As we have observed, there are clear differences between cohorts and this assumption is violated.

Furthermore, our data presents significant limitations for analysis, including incomplete coverage of total green investments, selection bias, and measurement errors due to its narrow focus on public sector funding. The extraction and processing of NZIP data encountered challenges such as missing information, assumptions on funding amounts, and location discrepancies, leading to an experimental dataset that captures less than half of the expected spend with a skewed regional distribution. These issues, coupled with potential biases from manual data handling and the inherent difficulties in measuring the impact of R&D investments, necessitate a cautious approach in interpreting the analysis results, highlighting the need for comprehensive data validation and consideration of broader investment sources to accurately assess the UK's green technology landscape.

7 Conclusion

In concluding our analysis on the impact of Green Investments, specifically through the NZIP, on UK productivity, it becomes evident that establishing a causal relationship between green investments and productivity enhancements presents significant challenges. The limitations and biases inherent in the available data, including the difficulty of controlling for spillover effects and the lack of comprehensive controls for other values of green investment, underscore the complexity of this task. Despite these challenges, our findings suggest a modest observational increase in labour productivity, ranging between £0.27 and £0.52, occurring 3-4 years post-NZIP funding in Local Authorities where businesses benefited from such investments. This indicates a potential positive impact of green investments on productivity, albeit within a constrained and specific context.

While not comprehensive, our analysis importantly does not reject the hypothesis that Green Investment reduces productivity, indicating the potential harmonious relationship. Our work also helps in identifying important characteristics to deploying net zero investment to partially resolve the UK's productivity problem. We have identified that there is likely heterogeneity in sectors, regions and time periods. We would suspect that regions with already low productivity are likely to benefit more from green investment than those with already high levels and that benefits may accrue over the long-term with slight negative impacts in the short. This highlights two key areas of policy for government, the potential dual benefits of levelling up low productivity regions and helping achieve net zero emissions by 2050, and the importance of promoting long-term investment by businesses.

Looking forward, the analysis could be enriched and its scope broadened by (1) incorporating additional productivity measures such as capital productivity and total factor productivity to capture the wider effects on green investment, (2) drawing international comparisons, and (3) applying more refined spatial HAC methodologies, such as Conley (1998), to better account for spillover effects. Moreover, identifying sources of exogenous variation, such as utilising near misses in investment funding as a basis for a regression discontinuity approach, could provide a more robust causal identification of the effects of green investments on productivity.

The overarching conclusion underscores a critical need for the government to make available high-quality, granular datasets on large-scale investment plans. Such data would not only facilitate more nuanced and insightful analyses but also enable a deeper understanding of the relationship between green investments and productivity enhancements. The potential insights gleaned from improved data availability could significantly inform policy decisions, guiding more effective and impactful investment strategies in the pursuit of sustainable economic growth and productivity improvements within the UK.

8 References

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9 ANNEX

9.1 Annex A

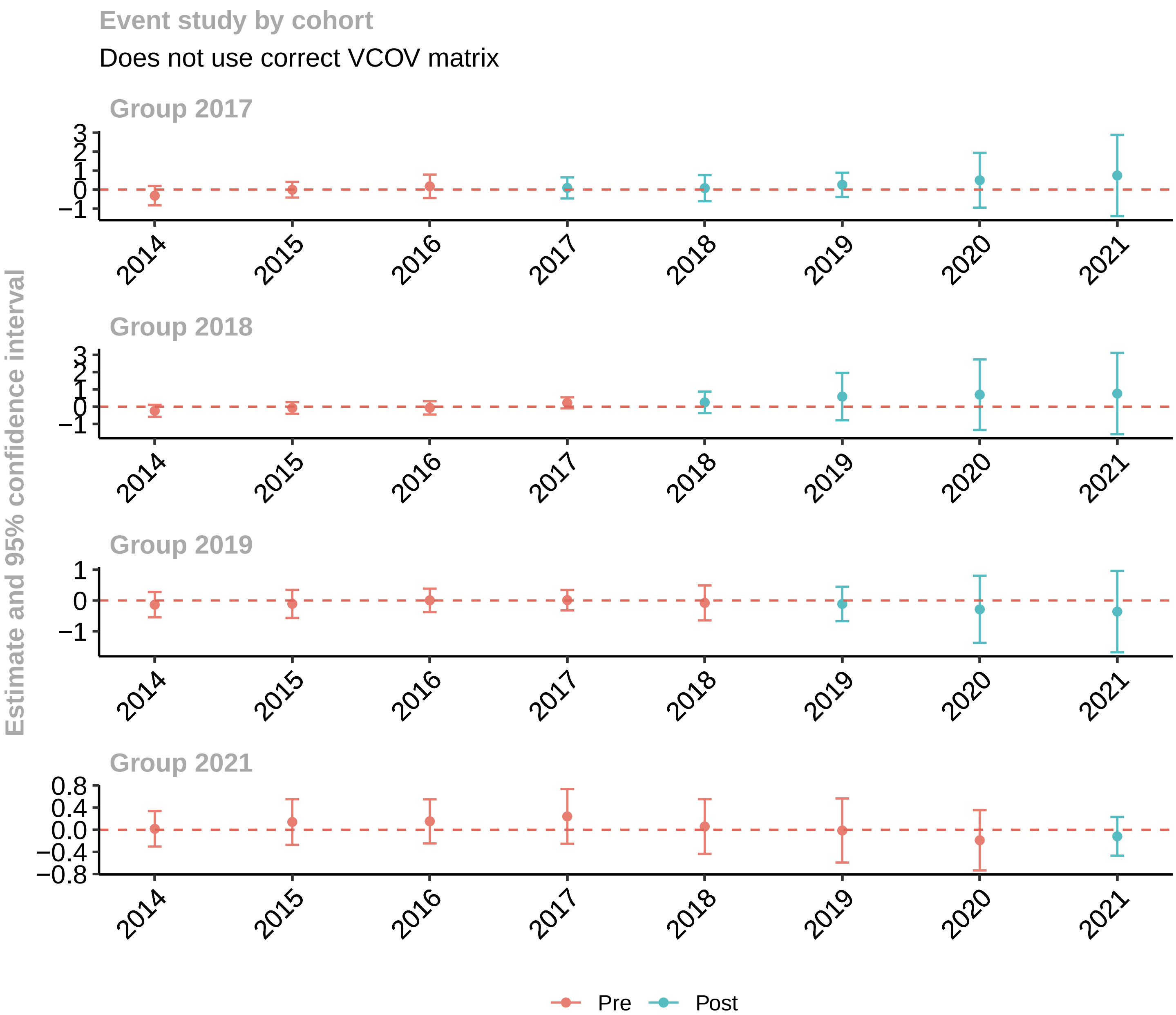

Figure 2: Event Study Test for Parallel Trends

N.B: This specification does not use DK Standard Errors and therefore confidence intervals may be exaggerated.

9.2 Annex B

Table 3: Hausmann test for Random vs Fixed effects

Metric	Value
Data	value_a3 ~ Total_Funding + treat + post + i(treat, post)
Chi-squared	5.2758
Degrees of Freedom	4
P-value	0.02602
Alternative Hypothesis	one model is inconsistent

9.3 Annex C

Table 4: Lagrange Multiplier test for conditional heteroskedasticity

Metric	Value
Test	Lagrange Multiplier Test - time effects (Breusch-Pagan)
Data	value_a3 ~ Total_Funding + treat + post + i(treat, post)
Chi-squared	3420.6
Degrees of Freedom	1
P-value	< 2.2e-16
Alternative Hypothesis	significant effects

9.4 Annex D

Table 5: Breusch-Pagan test for heteroskedasticity

Metric	Value
Test	studentized Breusch-Pagan test
Data	fixed
BP	58.549
Degrees of Freedom	4
P-value	5.851e-12

9.5 Annex E

Table 6: Goldfeld-Quandt test for conditional heteroskedasticity

Metric	Value
Test	Goldfeld-Quandt test
Data	fixed
GQ	0.91013
df1	1055

Metric	Value
df2	4189
P-value	0.9716
Alternative Hypothesis	variance increases from segment 1 to 2

9.6 Annex F

Table 7: Sun and Abraham Estimation Results

Variable	Coefficient	Standard Error
Dependent Var.:	**Productivity**	
Total_Funding	1.29e-8	(1.18e-8)
year = -5	-0.0941***	(7.7e-15)
year = -4	0.1461***	(8.72e-15)
year = -3	0.1714***	(5.49e-15)
year = -2	0.1061***	(9.38e-15)
year = 0	-0.0817*	(0.0290)
year = 1	0.0274	(0.0474)
year = 2	0.0675	(0.0474)
year = 3	0.6022***	(0.0510)
year = 4	0.6821***	(0.0567)
Fixed-Effects:		
lad_name	Yes	
year	Yes	
S.E. type:	Driscoll-Kraay (L=1)	
Observations:	1,752	
R2:	0.96727	
Within R2:	0.00359	
Signif. codes:	0.001 ‘***’ 0.01 ‘**’ 0.05 ‘*’ 0.1 ‘.’	